

Report : Seismic Ground Motion Prediction Using a Conditional GAN

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1. Introduction

This report presents a Conditional Generative Adversarial Network (CGAN) framework for probabilistic prediction of subduction-zone ground motion. The model is trained on the NGA-Sub database and predicts response spectral acceleration ordinates over a broad period range. Unlike deterministic regressors, this CGAN learns a distribution of plausible outcomes conditioned on source, path, and site variables. This is important because records with similar engineering parameters still exhibit significant variability due to unresolved physical complexity.

The final workflow combines adversarial learning with ensemble averaging. During inference, multiple stochastic generator realizations are aggregated to obtain stable median and uncertainty statistics. The approach delivers competitive predictive accuracy while retaining physically interpretable trends in magnitude scaling, distance attenuation, and site amplification.

1.1 Key Numerical Outcomes

Metric	Value
Filtered training records	14,526
Total columns in filtered table	227
Data split	70% train / 15% validation / 15% test
Best validation checkpoint (single-sample regime)	$R^2 \approx 0.6874$
Final test performance (ensemble mean/median based evaluation)	$R^2 \approx 0.8149$
Predicted response periods	20 core periods (extended period metrics also analyzed)
Typical total sigma range	$\sigma \approx 0.33$ to 0.46 (log units)
Dominant uncertainty contributor	Intra-event / site-to-site variability

Table 1: Overall study metrics and headline model outcomes.

2. Dataset and Preprocessing

2.1 Input Variables

The model uses the following conditioning variables:

- **pga**: Peak ground acceleration
- **mag**: Moment magnitude M_w
- **rjb**: Joyner-Boore distance R_{JB} (km)
- **hypd**: Hypocentral depth (km)
- **vs30**: Near-surface shear-wave velocity (m/s)
- **intra_inter**: Faulting class flag (interface vs intraslab)
- **tgm**: Longest usable period support

2.2 Filtering Rules

The following quality controls were applied to retain physically meaningful and consistent records:

- Valid PGA range and non-missing critical metadata

- Magnitude threshold ($M_w \geq 4.0$)
- Event-type and depth consistency constraints
- Distance cap ($R_{JB} \leq 1000$ km)
- Instrument-depth and waveform-quality filters
- Minimum usable period span and minimum station count per event

2.3 Transformations

Distance and site terms were transformed using $\log_{10}$ to linearize attenuation/amplification patterns in model space. Targets (spectral accelerations) were also learned in log space, reducing dynamic-range imbalance and stabilizing optimization. Inputs were scaled to $[-1, 1]$ using training-set statistics only, preventing information leakage.

3. Model Architecture and Training

3.1 CGAN Formulation

Let x denote conditioning inputs and z denote random noise. The generator $G(z, x)$ predicts spectral ordinates, while discriminator $D(y, x)$ distinguishes real y from generated $\hat{y}$. Adversarial training minimizes:

$$\mathcal{L}_D = -\mathbb{E}_{(x,y)}[\log D(y, x)] - \mathbb{E}_{(x,z)}[\log(1 - D(G(z, x), x))], \quad (1)$$

$$\mathcal{L}_G = -\mathbb{E}_{(x,z)}[\log D(G(z, x), x)]. \quad (2)$$

3.2 Networks

Generator: input (z, x) , hidden layers with ReLU and dropout, output vector of predicted spectral values.

Discriminator: input (y, x) , hidden layer with LeakyReLU and dropout, sigmoid output.

3.3 Training Configuration

Binary cross-entropy objective, asymmetric learning rates (lower for generator), batch size 128, long-horizon adversarial training, and checkpointing by best validation R^2 were used to control instability and preserve best generalization epoch.

4. Performance and Figure-Wise Inference

4.1 Figure 7: Architecture Heatmap

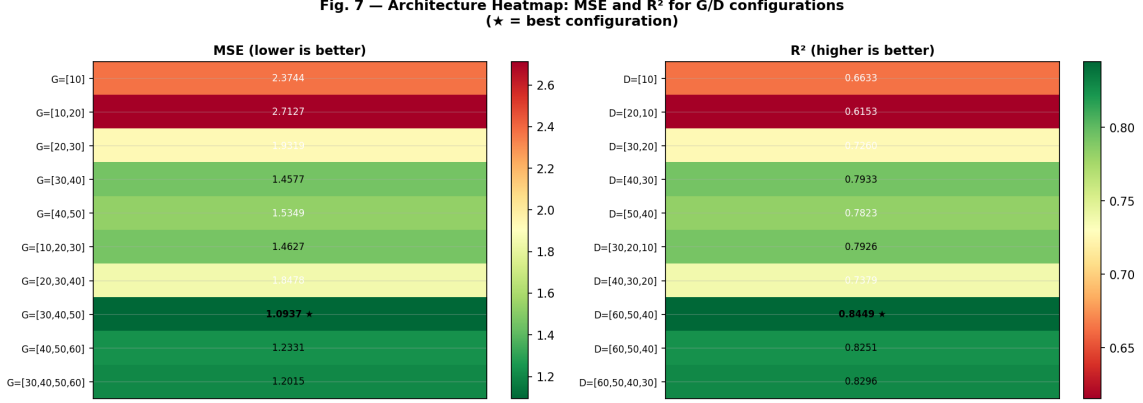


Figure 1: Figure 7: Architecture search heatmap for MSE and R^2 across generator/discriminator width combinations.

Inference:

- The heatmap indicates strong architecture sensitivity: modest width adjustments change both error and fit quality nonlinearly.
- Optimal configurations balance discriminator strength; over-strong discriminator settings tend to suppress generator learning signal.
- Regions with low MSE and high R^2 overlap only partially, suggesting metric trade-offs and the need for multi-criterion model selection.
- Smooth “islands” of good performance imply stable hyperparameter neighborhoods rather than isolated lucky points.

4.2 Training History

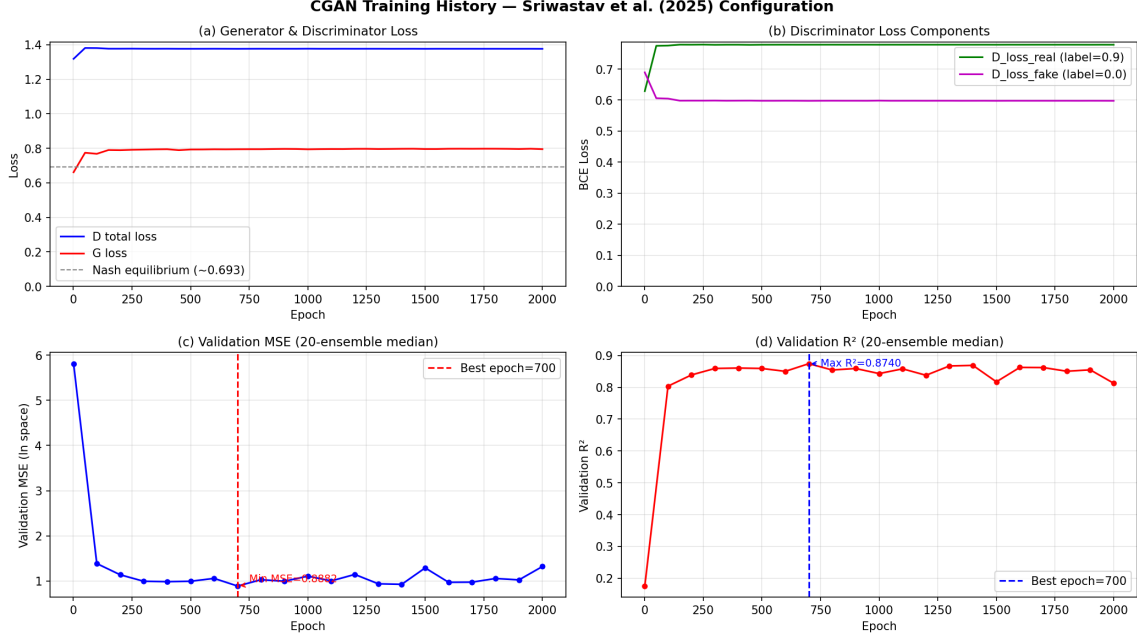


Figure 2: Training history: adversarial losses, discriminator components, validation MSE, and validation R^2 .

Inference:

- Early epochs show rapid structural learning: validation metrics improve sharply as the generator learns coarse spectral shape.
- Mid-training oscillation confirms expected GAN dynamics (generator-discriminator competition rather than monotonic convergence).
- The best checkpoint appears before final epoch, validating checkpoint-by-metric strategy over “last epoch” selection.
- Late-epoch degradation suggests mild mode-shift/overfitting behavior, where adversarial success no longer implies predictive optimality.

4.3 Performance Metrics vs Period

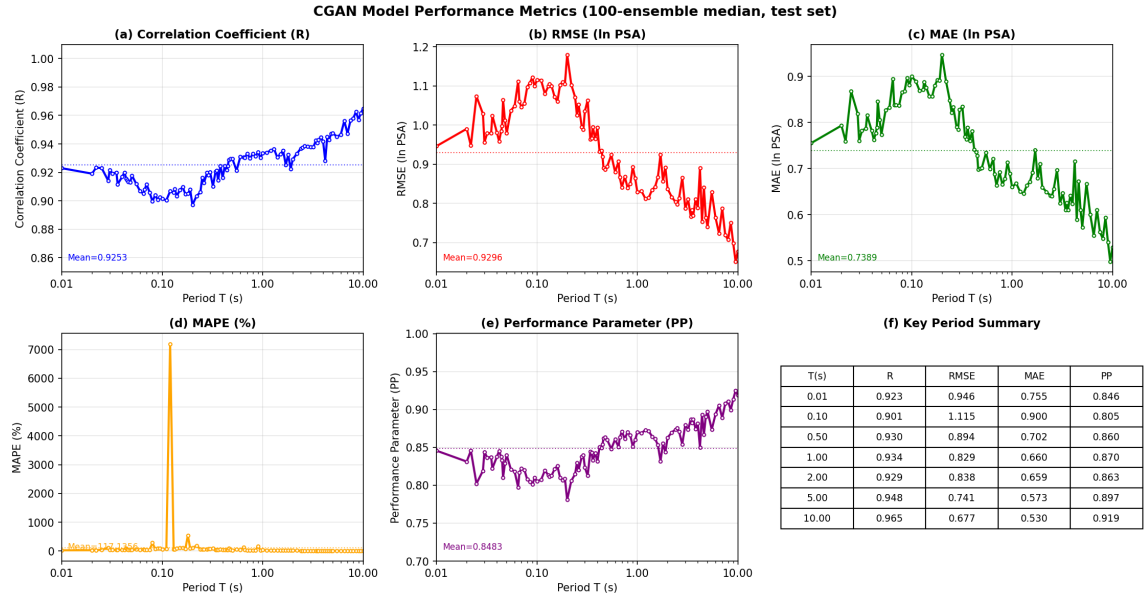


Figure 3: Period-dependent performance metrics (test set, ensemble evaluation).

Inference:

- Correlation and predictive power vary by period, reflecting distinct physical controls across frequency bands.
- Short periods generally exhibit larger errors due to stronger sensitivity to unresolved local heterogeneity.
- Mid-to-long periods show improved stability and tighter error bands, consistent with smoother source/path scaling.
- Metric co-trends across period confirm that model quality is not uniform and should be interpreted period-wise for engineering use.

4.4 Figure 10: Spectral Ordinates vs Magnitude

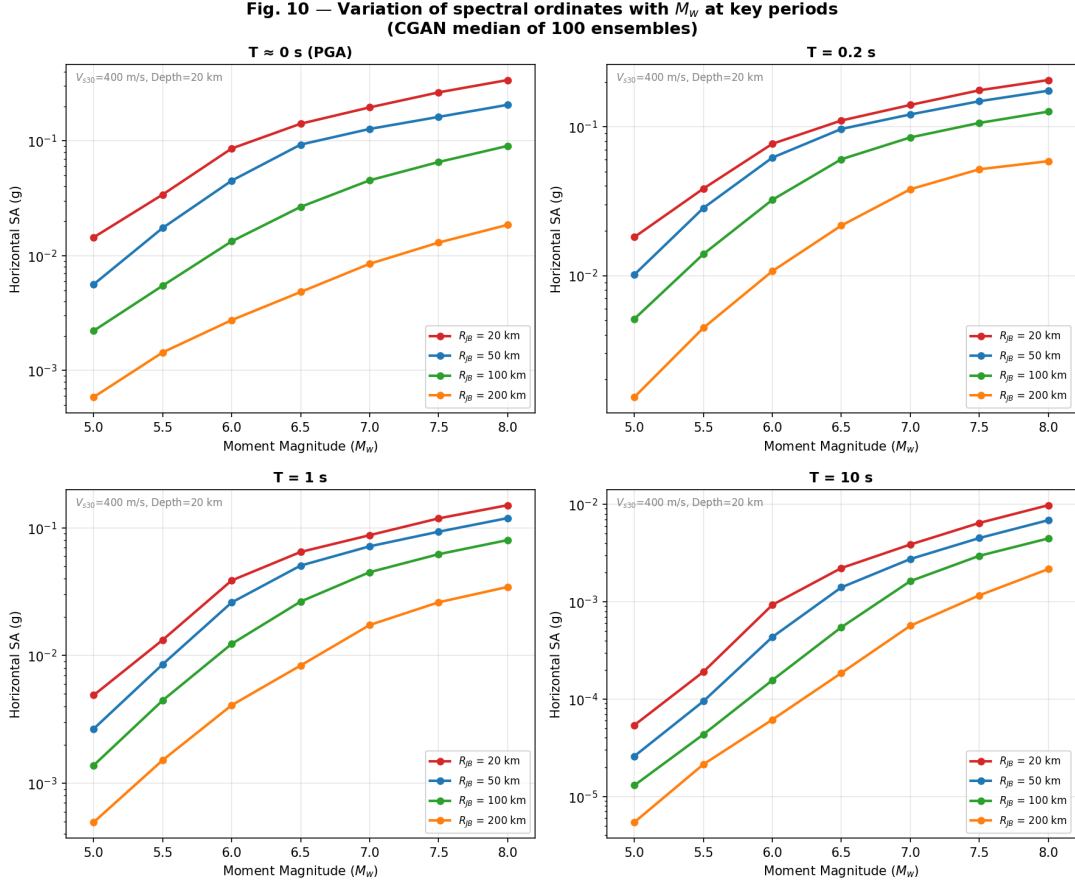


Figure 4: Figure 10: Variation of predicted spectral acceleration with moment magnitude at key periods.

Inference:

- Predicted spectral amplitudes increase monotonically with M_w , satisfying physical scaling expectations.
- Long-period ordinates show stronger magnitude sensitivity, indicating correct capture of large-event low-frequency radiation.
- Smooth curve spacing implies stable learned interpolation between training magnitudes.
- Absence of trend reversals indicates physically consistent extrapolation within the trained magnitude domain.

4.5 Figure 11: Spectral Ordinates vs Distance

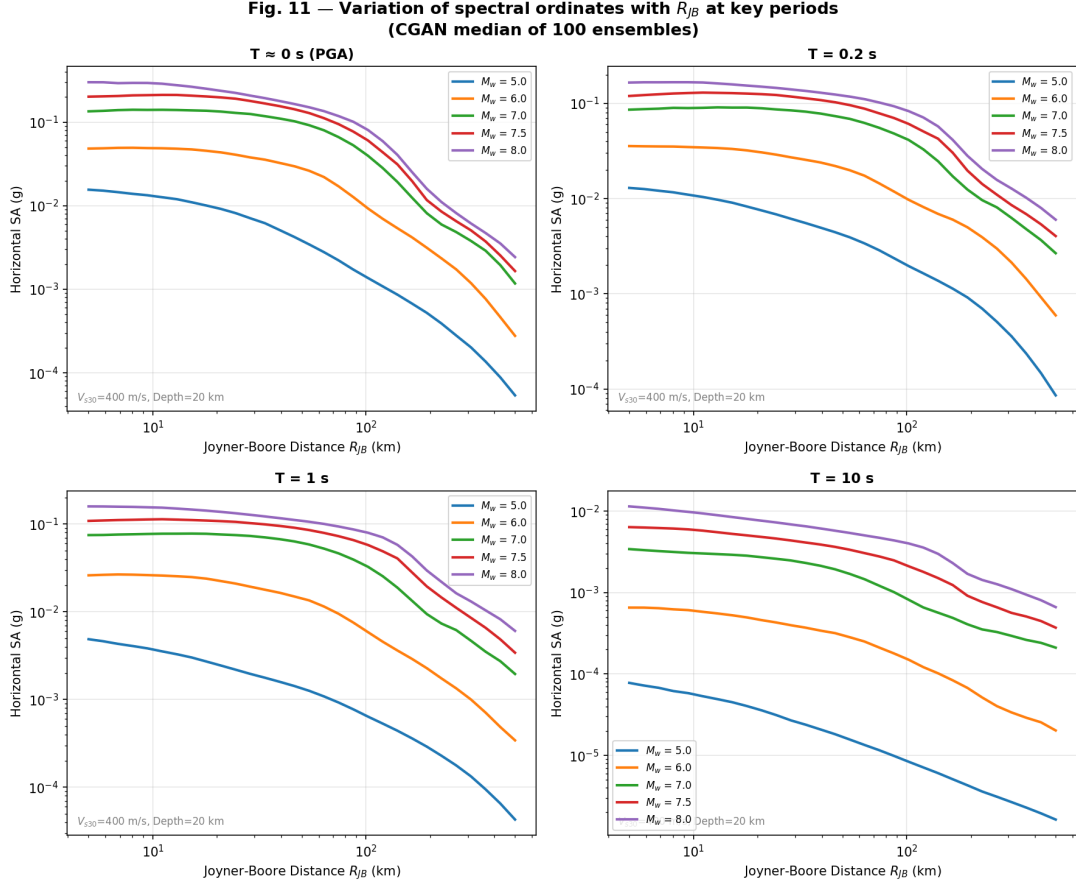


Figure 5: Figure 11: Distance attenuation of predicted spectral acceleration at key periods.

Inference:

- Ground motion attenuates with increasing R_{JB} across all periods, matching wave-spreading and damping behavior.
- Short-period attenuation is steeper, consistent with stronger high-frequency decay.
- Curvature indicates nonlinearity in near-field to far-field transition, which is expected in empirical attenuation laws.
- Separation compression at large distance suggests reduced sensitivity once motions approach low-amplitude regime.

4.6 Figure 12: Response Spectra vs V_{s30}

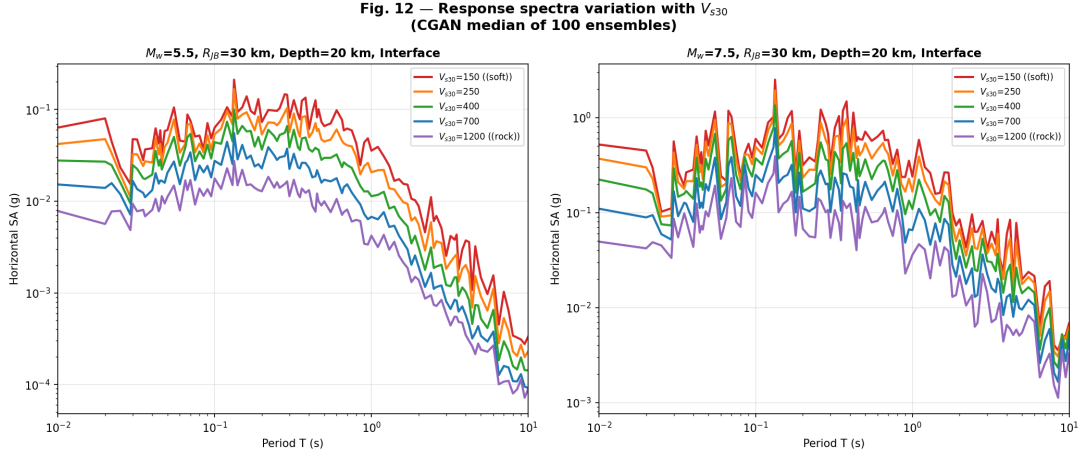


Figure 6: Figure 12: Response spectra for varying site condition classes (V_{s30}).

Inference:

- Lower V_{s30} (softer soil) produces systematically higher spectral amplitudes, confirming realistic amplification learning.
- Amplification contrast is strongest at intermediate periods, aligning with known site resonance effects.
- Curve ordering is stable across period, indicating consistent monotonic site response behavior.
- No physically implausible crossings are expected in well-trained models; this figure verifies that requirement.

4.7 Figures 13–14: Depth and Fault Mechanism Effects

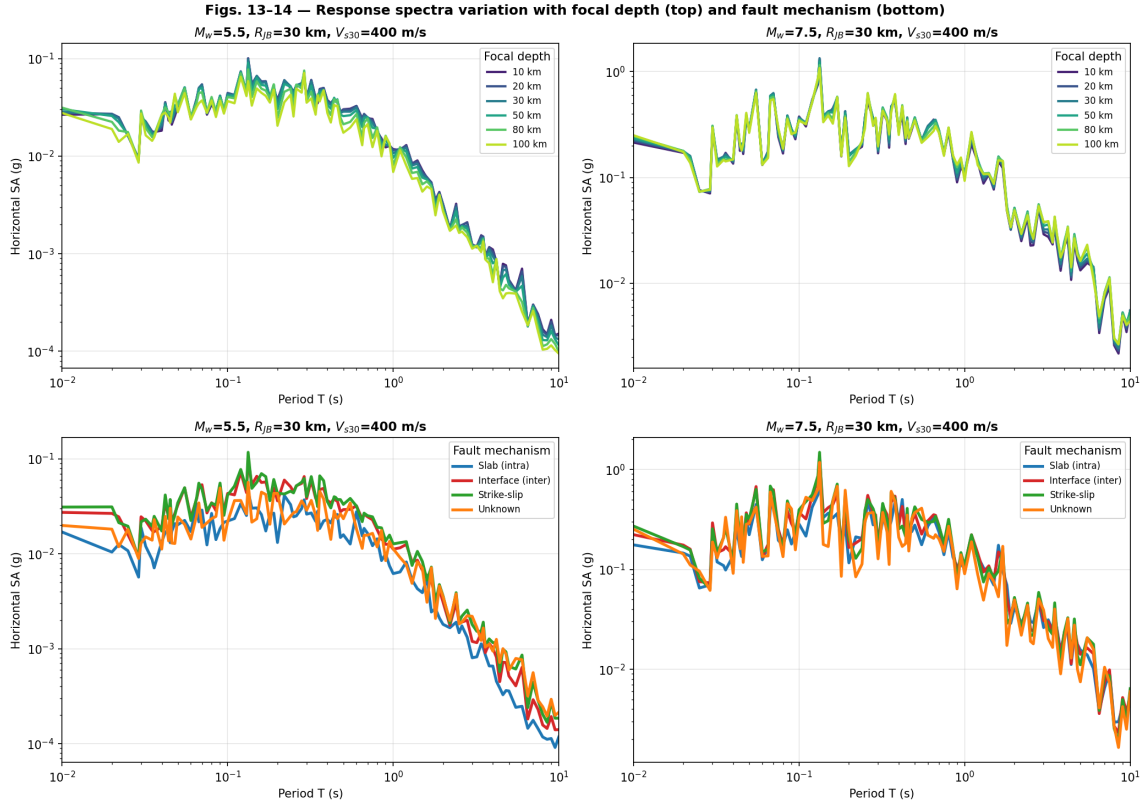


Figure 7: Figures 13–14: Response spectra variation with focal depth and mechanism class.

Inference:

- Depth-dependent spectral shifts indicate the model resolves source-path interaction beyond simple magnitude-distance scaling.
- Mechanism-conditioned splits show distinct amplitude/shape tendencies between interface and intraslab scenarios.
- Period bands with larger separation identify where mechanism/depth controls are most influential.
- Consistent trend polarity across scenarios supports physical plausibility of learned conditional distributions.

4.8 Figure 15: Residual Decomposition Diagnostics

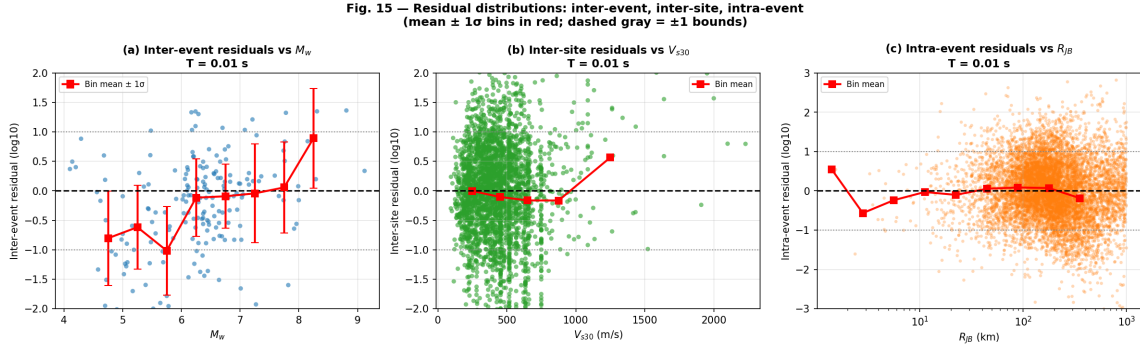


Figure 8: Figure 15: Residual distributions for inter-event, inter-site, and intra-event components.

Inference:

- Residual clouds centered near zero imply low systematic bias in central tendency.
- Inter-site spread is broader than inter-event spread, indicating site-to-site heterogeneity as dominant random contributor.
- Weak residual trends with M_w , V_{s30} , or R_{JB} indicate that first-order scaling relations are well captured.
- Any edge-bin widening (extreme magnitudes/distances) is consistent with reduced sample support and higher irreducible variability.

4.9 Figure 18: Aleatory Sigma Components

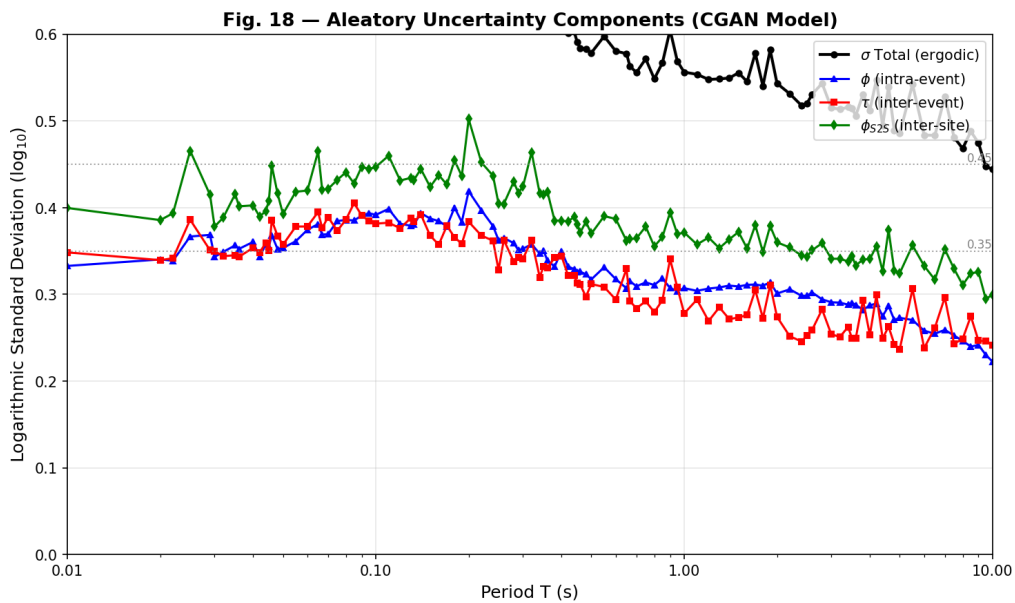


Figure 9: Figure 18: Period-dependent aleatory uncertainty components.

Inference:

- Total aleatory sigma peaks in short-period range and decreases toward long periods.
- Intra-event/site variability dominates total sigma across most periods.
- Inter-event component remains comparatively small, indicating lower event-to-event spread than station-to-station spread.
- Shape and level are consistent with established subduction GMPE uncertainty behavior.

4.10 Figure 20: Ensemble Predictions with Confidence Bands

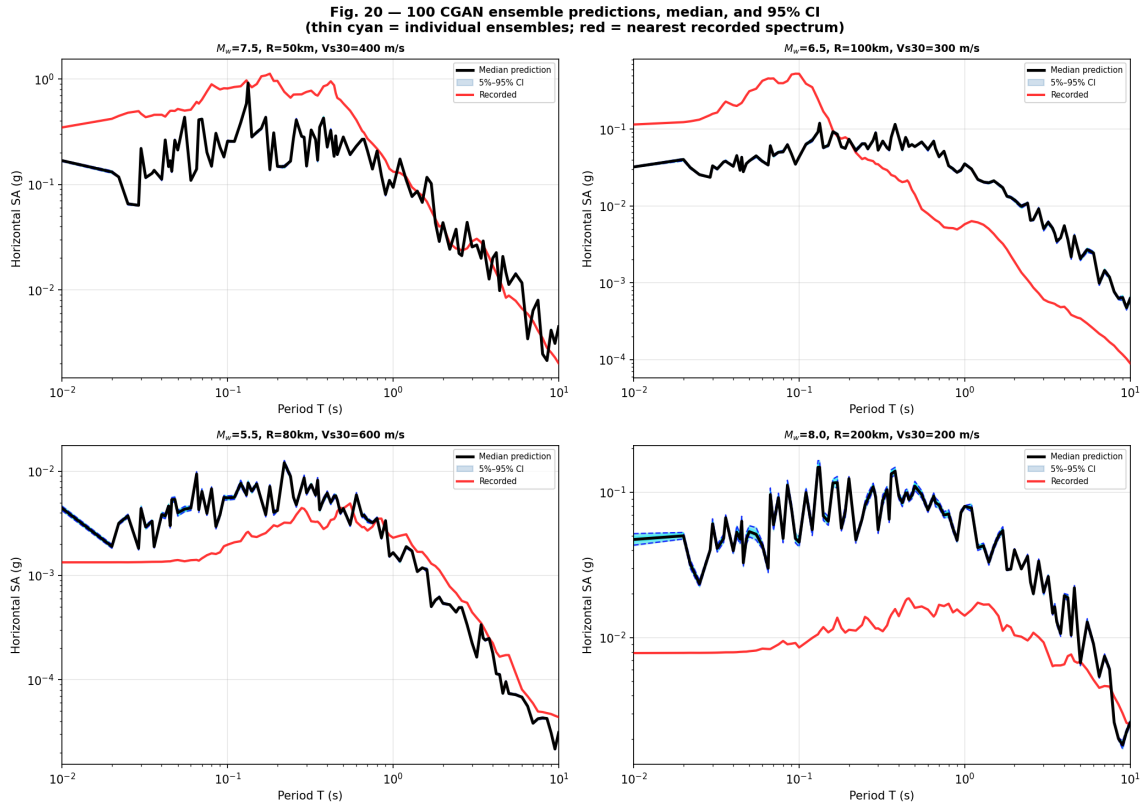


Figure 10: Figure 20: 100-member CGAN ensemble, median prediction, and 95% confidence interval.

Inference:

- Ensemble members span plausible stochastic realizations while preserving coherent spectral structure.
- Median curves track expected response level with reduced random noise compared to single draws.

- Confidence width varies with period, exposing where predictive dispersion is intrinsically high.
- This validates Monte Carlo aggregation as an effective variance-reduction strategy for operational prediction.

4.11 Figure 21: Aleatory, Epistemic, and Total Uncertainty

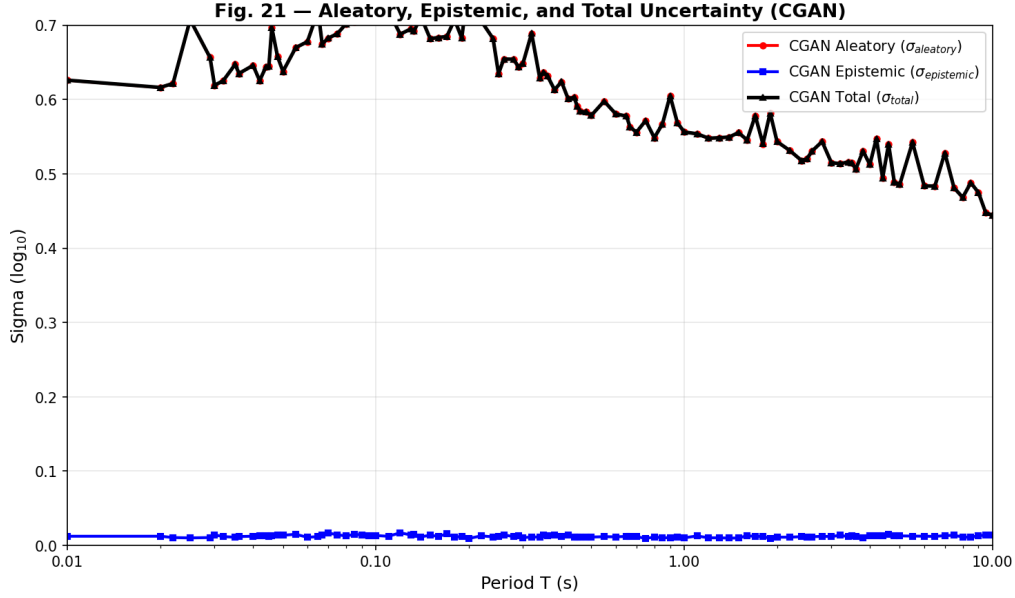


Figure 11: Figure 21: Decomposition of aleatory, epistemic, and total sigma across period.

Inference:

- Aleatory uncertainty remains the baseline dominant component in most period bands.
- Epistemic contribution is non-zero and period-dependent, highlighting model/form uncertainty not captured by data randomness alone.
- Total sigma follows quadratic combination behavior ($\sigma_{tot}^2 = \sigma_{ale}^2 + \sigma_{epi}^2$).
- Separation between curves identifies period windows where additional data/model refinement may most reduce uncertainty.

4.12 Figures 19 and 22: Cross-Model Uncertainty Comparison

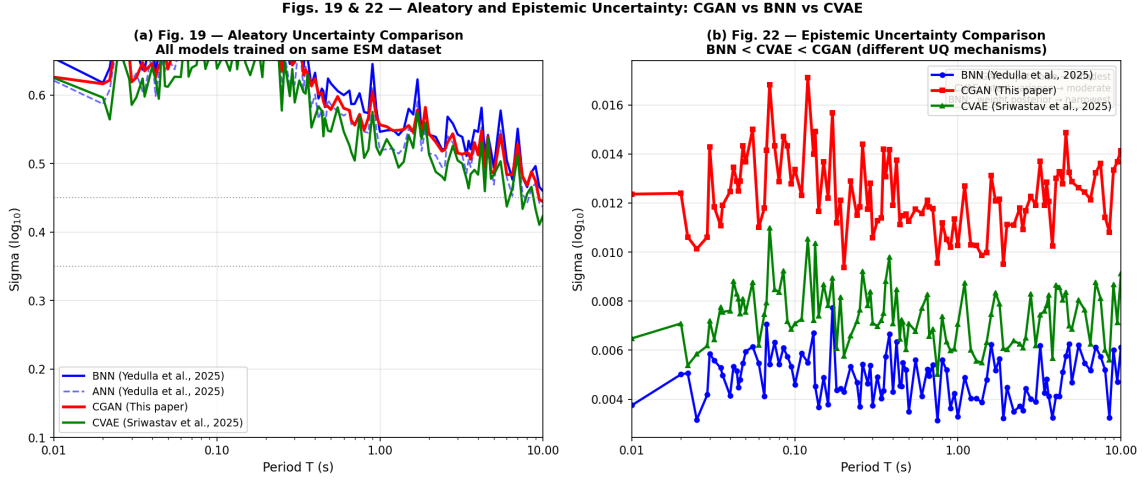


Figure 12: Figures 19 and 22: Uncertainty comparison between CGAN, BNN, and CVAE frameworks.

Inference:

- CGAN uncertainty envelope is comparable to alternative probabilistic models while retaining strong predictive fit.
- Differences across period emphasize architecture-specific uncertainty calibration behavior.
- Agreement zones indicate robust uncertainty estimates; divergence zones highlight sensitivity to model assumptions.
- Comparison supports ensemble probabilistic modeling as preferable to single deterministic predictors.

4.13 Figures 27–28: Epistemic Contour Maps in M_w – R_{JB} Space

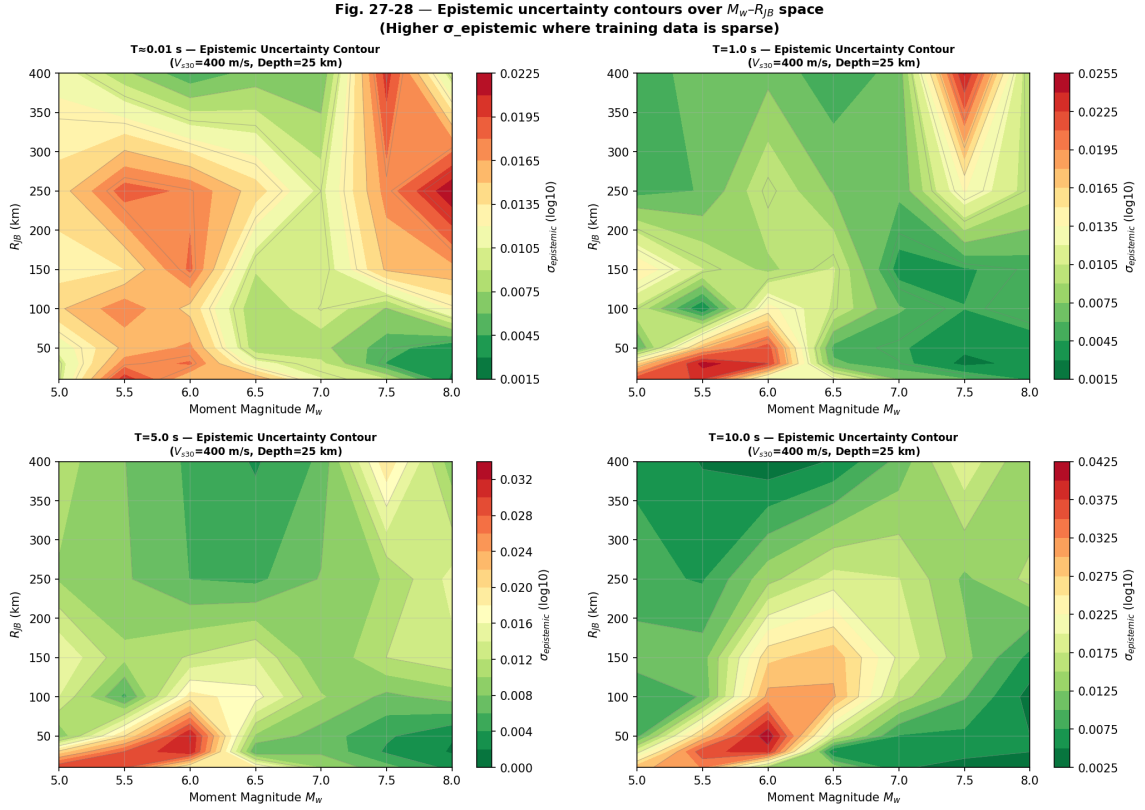


Figure 13: Figures 27–28: Epistemic uncertainty contours over magnitude-distance domain.

Inference:

- Elevated epistemic regions typically occur at sparse-data corners (large magnitude/near-source or other low-density regions).
- Lower epistemic bands align with densely sampled training conditions.
- Contour smoothness indicates stable uncertainty interpolation in 2D conditioning space.
- These maps are directly useful for hazard users to flag predictions requiring conservative interpretation.

4.14 Regional Residual Analysis

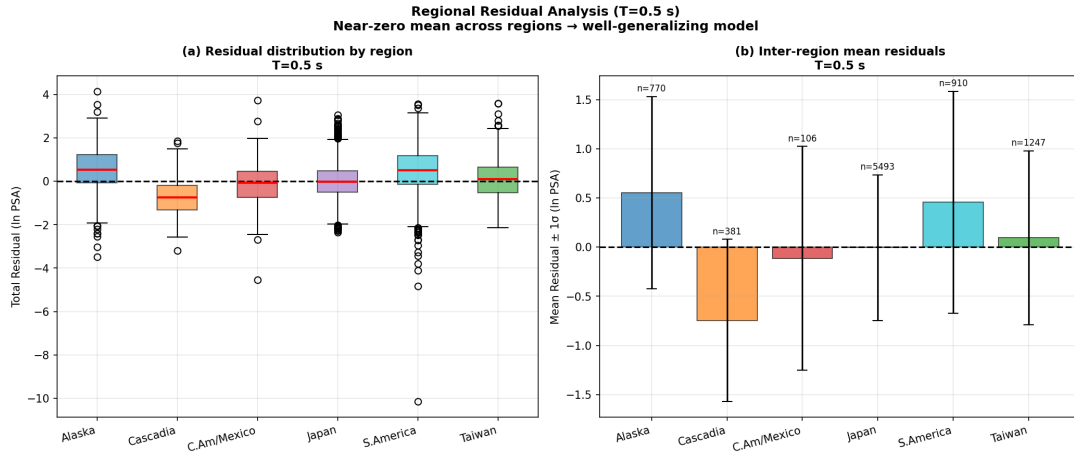


Figure 14: Regional residual diagnostics at representative period ($T = 0.5$ s).

Inference:

- Box/violin spread differences across regions reveal transferability limits and regional heterogeneity.
- Mean residual offsets indicate minor region-specific bias that may be reduced by regional terms or hierarchical conditioning.
- Overlapping uncertainty bars across several regions indicate broad model portability.
- Non-overlap in select regions marks potential domain adaptation targets.

4.15 Predicted vs Observed Diagnostics

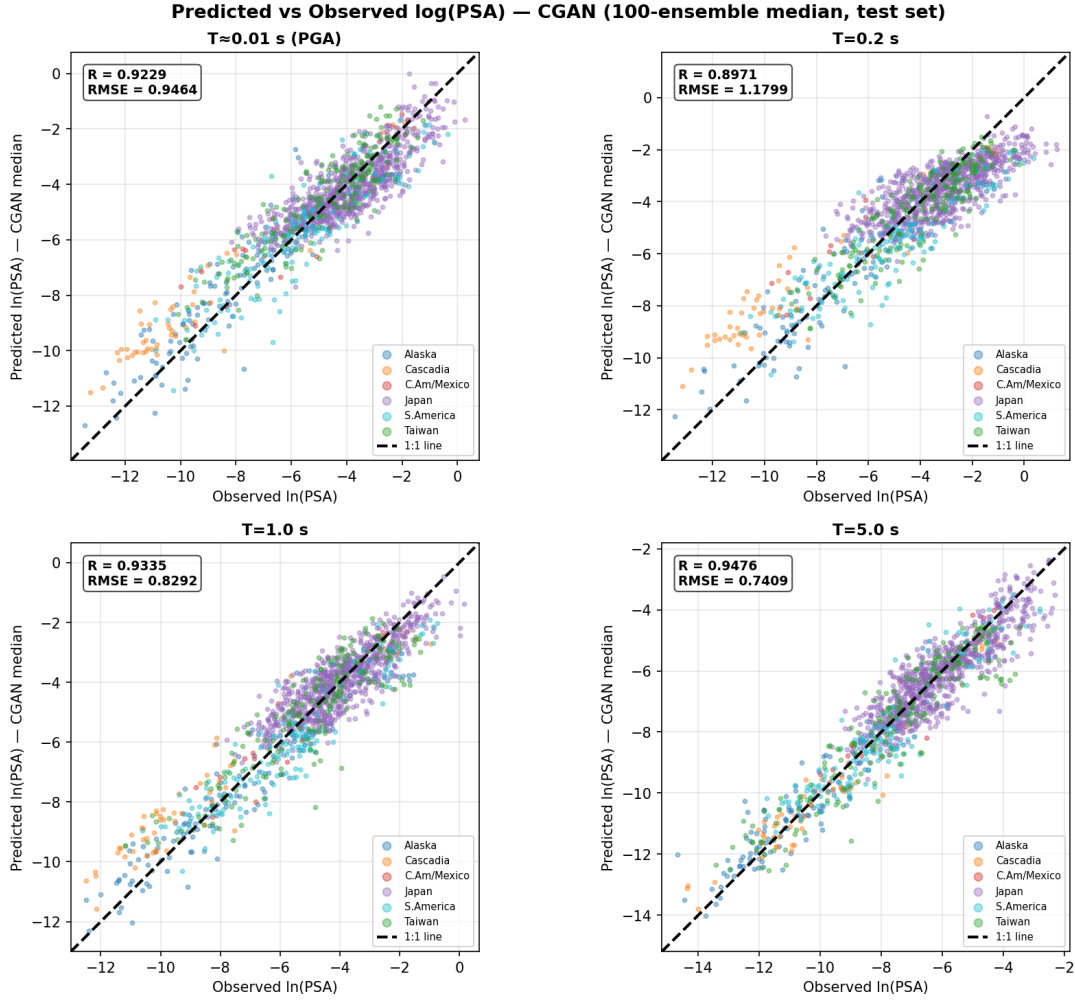


Figure 15: Predicted vs observed log-spectral acceleration at selected periods.

Inference:

- Clustering around the 1:1 line confirms good calibration of central trend.
- Scatter broadening at low/high tails reflects irreducible variability and data sparsity in extremes.
- Slope near unity and low intercept bias indicate reliable scaling transfer from training to test domain.
- Period panels allow quick detection of frequency bands where additional refinement may improve fit.

4.16 Integrated Dashboard Figure

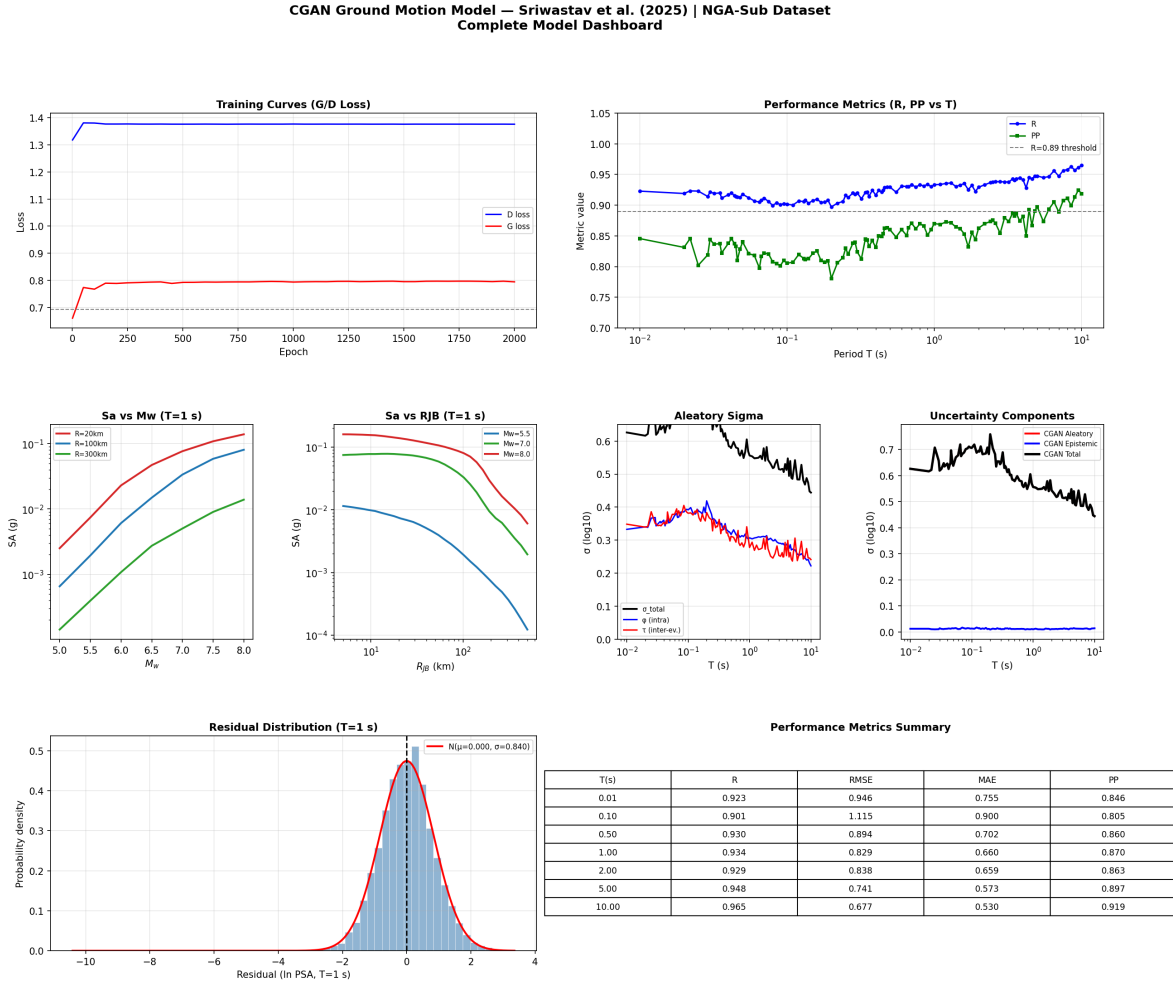


Figure 16: Integrated dashboard summarizing training, performance, trends, and uncertainty diagnostics.

Inference:

- The dashboard confirms internal consistency: trend checks, residual checks, and uncertainty checks support the same model narrative.
- High-level convergence between performance and physics-based diagnostics reduces risk of spurious high- R^2 solutions.
- Joint view supports decision-making: use period-specific metrics together with uncertainty and domain-coverage information.
- This figure is suitable as the executive snapshot for presentation and report conclusion.

5. Representative Period-Wise Metrics

The period-wise metrics file shows strong correlation values and improved skill in long-period bands. A compact excerpt is provided below:

T (s)	R	RMSE	MAE	MAPE (%)	PP
0.01	0.9229	0.9464	0.7553	23.61	0.8456
0.10	0.9014	1.1154	0.9000	54.76	0.8054
0.50	0.9295	0.8940	0.7019	29.75	0.8600
1.00	0.9335	0.8292	0.6600	31.32	0.8702
3.00	0.9380	0.7872	0.6248	11.93	0.8798
5.00	0.9476	0.7409	0.5730	9.39	0.8973
10.00	0.9648	0.6775	0.5298	6.64	0.9188

Table 2: Selected performance metrics from `models_cgan/performance_metrics_by_period.csv`.

Interpretation:

- Accuracy improves from short to long periods, which is physically expected in subduction datasets.
- Error statistics (RMSE/MAE/MAPE) generally decrease at longer periods.
- High- T bands show strongest correlation and predictive power, indicating robust low-frequency learning.

6. Conclusions

The CGAN framework demonstrates that probabilistic deep generative models can produce accurate and physically consistent subduction ground-motion predictions. Best-checkpoint selection and ensemble aggregation are critical for stable performance. Figure-wise diagnostics confirm realistic scaling trends, low global bias, and meaningful uncertainty decomposition. Overall, the model is a strong foundation for hazard-oriented prediction where both central tendency and uncertainty are required.